WASP-3b

R-0276

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_170508` · created 2026-07-17T14:20:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457881.8428 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.041
Transit depth (Rp/Rs)^2	2.04 +/- 1.18 [%]
Orbital Inclination (inc)	87.55 +/- 2.87
Airmass coefficient 1 (a1)	0.56 +/- 0.27
Airmass coefficient 2 (a2)	0.28 +/- 0.15
Scatter in the residuals of the lightcurve fit is	48.33 %
Best Comparison Star	None
Optimal Aperture	5.83
Optimal Annulus	46.63
Transit Duration (day)	0.0871 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.041 vs 0.10538 (deviation 0.3σ)
Duration fit vs published	0.0871 d vs 0.1167 d (deviation 1.63σ)
Mid-time O–C	-3.9 min
Depth SNR	1.2
Residual scatter	48.33 %

Light curve

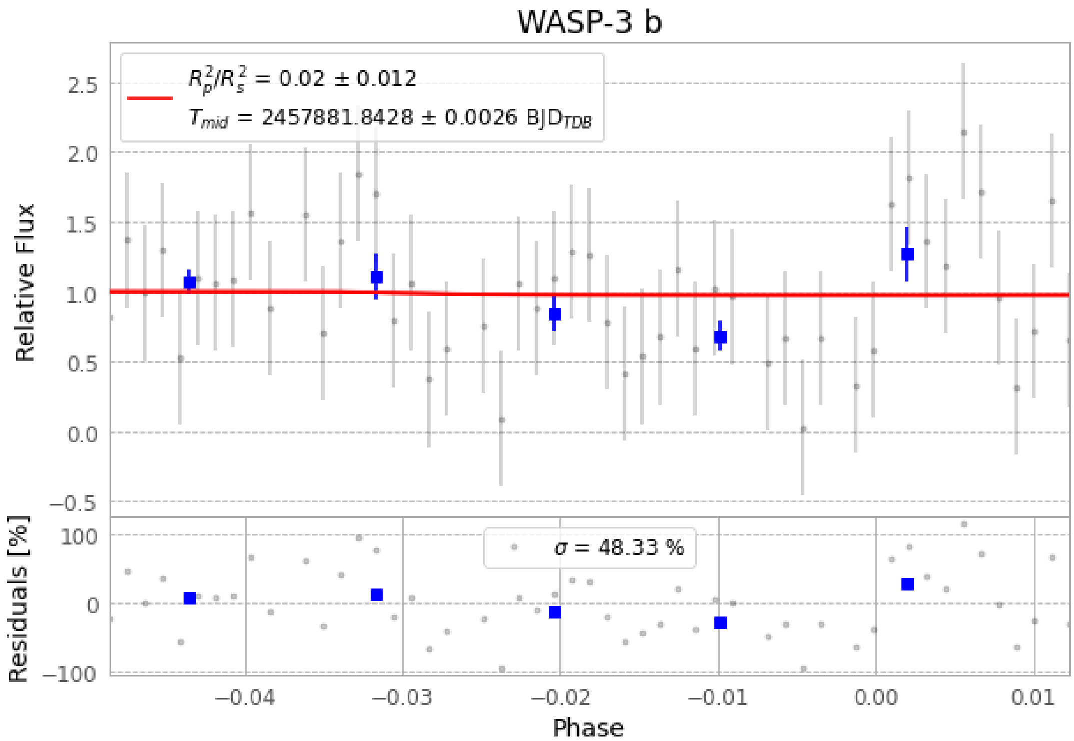

Figure 1 — FinalLightCurve_WASP-3 b_2017-05-07.png (SHA-256 f40dab70425c9c02…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 180], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7c631ca3410df727…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984`

Data provenance

54 FITS frames (35 MB), `WASP-3170508060012.FITS` → `WASP-3170508084212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-170508.log (57c699d72f8d…), FinalLightCurve_WASP-3 b_2017-05-07.png (f40dab70425c…), FinalLightCurve_WASP-3 b_2017-05-07.csv (eba0c6dd9233…)

Compute resources

Wall time 776.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-17 14:20Z.